

Artificial Intelligence and Rule-Based Reasoning Systems:

About Truffle Hunt: This exercise will allow you to use Artificial Intelligence to play a game. The game is *Truffle Hunt*, which is based on the classic game *Minesweeper*. You are searching for *truffles* – highly prized subterranean mushrooms that are buried in a field, and must be harvested carefully. You are accompanied by a *truffle-sniffing boar*¹, which is remarkable at locating truffles, but will eat them if it actually gets close enough to them. The game ends when you either isolate all 10 truffles (you win), or accidentally probe a truffle (you lose, because your boar eats the truffle). You may have failed, but at least your boar will love you.

You will provide a set of *rules* to train the boar, which accordingly examines the field and either makes a move or refrains from doing so. The two types of move a boar can make are (a) a *probe* in a cell deemed not to contain a truffle or (b) a *mark* in a cell deemed to contain a truffle. When it probes an empty cell, it is able to tell how many truffles are in the cells that surround it.

Depending on the rules you have used to train your boar, it might be able to deduce other cells that can safely be probed, or cells that are certain to contain a truffle, and can be marked. If the boar can find no place about which it feels certain, it refrains from moving. You may assume that in addition to their remarkable sniffing abilities, these boars speak flawless English.

Play the Game: Visit *Truffle Hunt* (<https://www.cs.drexel.edu/~popyack/TruffleHunt/index.html>). You will see a 10x10 grid, depicting the field. All cells are blank, indicating that we don't know anything about where the truffles are. You begin by clicking a cell, which instructs your boar to probe it. The cell will immediately display either an integer showing the number of neighboring cells that contain a truffle, or a red "X", designating that the boar actually found a truffle and ate it. (Game Over!) If the cell does not contain a truffle, the boar may decide to probe or mark another cell on its own, using an applicable rule from the Rule Center. If you select "Auto Boar", the boar will continue applying rules until it cannot find any more that apply.

This process continues, with you and the boar working together to isolate truffles. When you find a cell you feel certain contains a truffle, you can *right-click* on it (or *ctrl-click* it), and it will be marked in red. The boar likewise marks cells in red.

Understanding Rules: So how do you prescribe rules? Rules are in the form of *patterns* and *actions*. A pattern represents a 3x3 grid of symbols describing a configuration that might occur somewhere on the board. For instance, the data file currently contains two lines, representing a pattern and its associated action as described below:

Given a pattern **abcdefghi** which corresponds to the 3x3 matrix below, the possible pattern values are:

a	b	c
d	e	f
g	h	i

- a digit **0-8** meaning the specific value 0-8
- **N** means any *nonzero* digit
- **n** means *any* digit, including 0 (i.e., this cell has been explored and it is empty)
- **—** means blank (unexplored)
- **?** matches any character - including "out of bounds"
- **m** means the cell is "MARKED"

At least one of the cells in the pattern should be blank (unexplored), so that an action may be performed on it.

¹ Yes, Truffle-Sniffing Boars are real! See <https://www.facebook.com/watch/?v=3784215041864168> for instance.

